

# **A BLE-Based Bus Boarding System for the Visually Impaired**

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- **Barriers to Bus Use Among the Visually Impaired**

| Reason                                        | Percentage (%) |
|-----------------------------------------------|----------------|
| Difficulty identifying the desired bus number | 24.5           |
| Difficulty locating boarding and exit doors   | 8.4            |

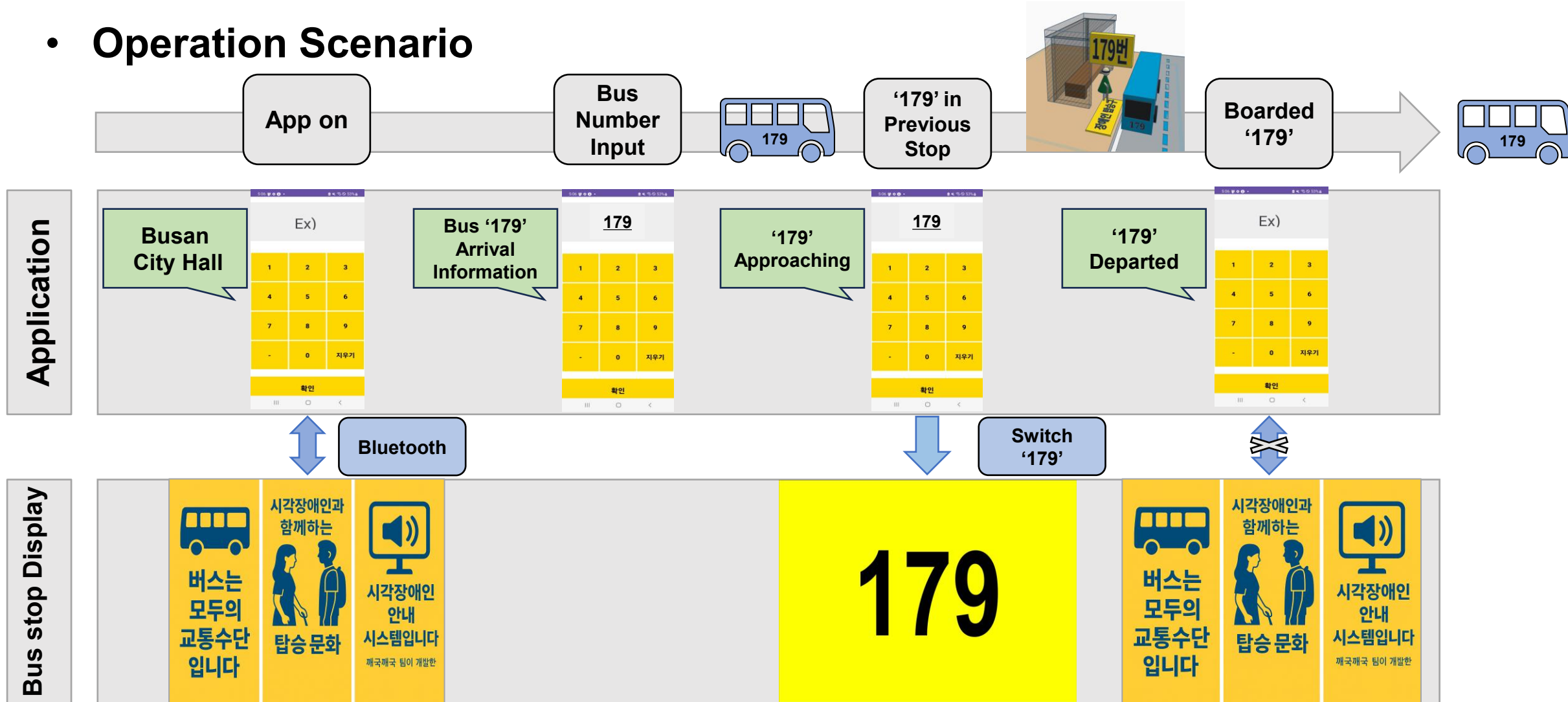
\*Source: Wello, Korea Blind Union

- **Solutions**

- Voice-guided application for bus number identification
- Bus stop display with a designated boarding zone

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- Operation Scenario



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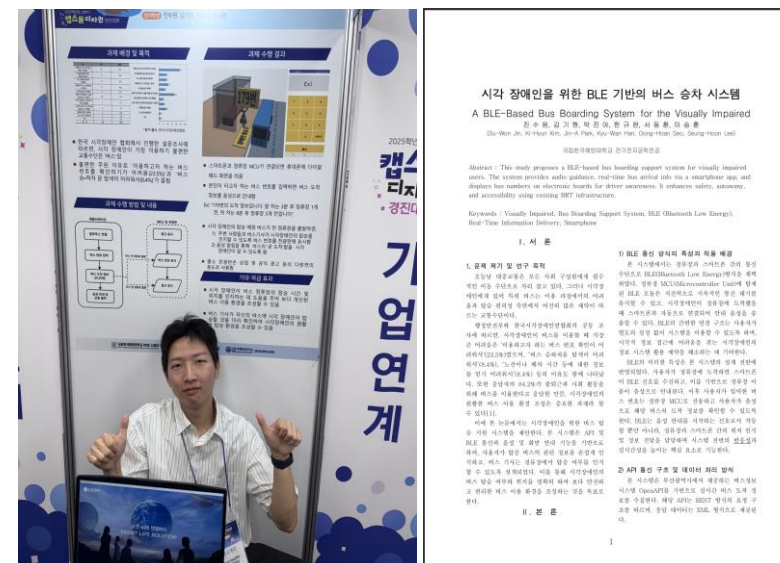
## • Results

2025.03 ~ 2025.06

- BLE-based app-display bus boarding system
- Competition & Conference participation
- Gold prize in departmental competition

## • My Contribution

- OpenAPI integration & XML parsing
- Event-driven display/alert logic
- Team Leader
- Paper preparation & submission
- Poster presentations at competitions and a conference



< Competition >

< Paper >